

Software Automation Testing Using Sikuli

Sikuli is a scripting language that can carry out automated software testing of graphical user interfaces (GUI) using screenshot images of the software under test.

Testing of any software project is as important as its development, and is done to check or validate different aspects like functional testing, security testing and database testing. The testing process can be manual or automated. Manual testing is performed by a person sitting in front of a computer, carefully executing tedious and time-consuming tests. Testing can also be done by using suitable automation tools, which makes it more reliable and faster. Whether one opts for manual or automated testing depends on various factors like the project's requirements, the budget, timelines, the expertise available and suitability.

A few major reasons why one should opt for automation testing are listed below:

- Manual testing is very time consuming when it comes to overall flow testing and covering all scenarios. Regression testing also becomes a very tedious task when done manually as it needs repetition of the same actions/steps.
- Manual testing becomes physically tiring.
- Manual testing is also less thorough than automation.
- To err is human; so manual testing is more error prone.

Automation testing may be useful in some cases but, sometimes, it may be too high-tech and could wind up costing you way more than it's worth. So it becomes very important to choose the correct automation tool for your project. The open source tools available include Selenium, Robotium, Autoit, Sahi and Sikuli, while some others are

licensed like HP Unified Functional Tool (UFT) and Tosca, for which one has to pay to use. Choosing open source tools can reduce the project cost; however, paid tools have many more features, are less time consuming and have wonderful support teams. So depending upon your project's requirement, you may opt for any automation tool to enhance your testing scope and speed.

Let's now discuss an emerging automation tool called Sikuli.

Introducing Sikuli

Sikuli is an open source automation tool that uses image recognition to identify and control GUI components. It can be integrated with the Selenium Web driver to automate Flash content and Java applets.

According to the official site, in the Huichol Native American language, Sikuli refers to God's Eye, implying the power to see and understand things unknown. It is basically a software framework licensed under MIT2.0 and is cross-platform. It was started in mid-2009 as an open source project by Tom Yeh and Tsung-Hsiang at the User Interface Design Group in MIT, USA. Both developers worked with Sikuli till Sikuli-X-1.0rc3 in 2012. Then Raimund Hocke (aka RaiMan) took over development support for Sikuli and maintained it. He developed it further as the SikuliX (where X denotes experimental) package together with the open source community, and continues to maintain it with its help.

